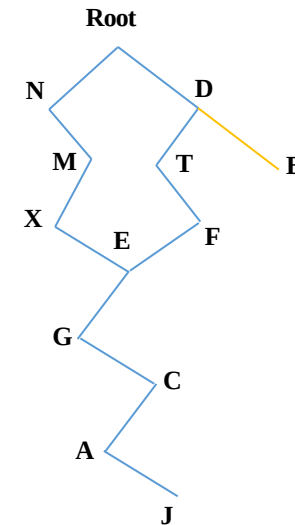


File Tagger Tool

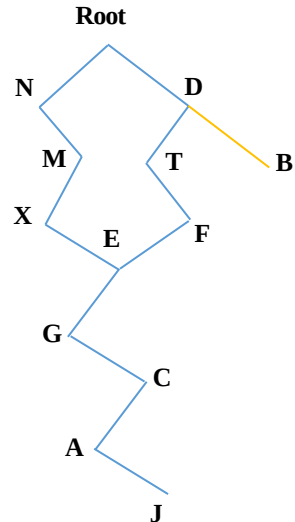
- Tool useful for bioinformatics labs to find user's files without knowing their names

Key Features

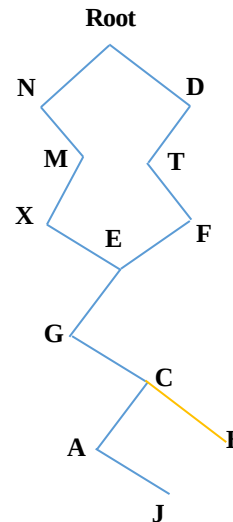
- Hierarchical Node Structure
- Tagging Files
- Command-Line Interface



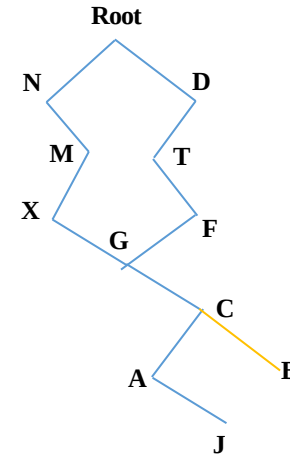
Hierarchical Node Structure



add nodes



Remove node



Commands: add and remove

```
file-tagger-main$ tagger.sh add -n N
file-tagger-main$ tagger.sh add -n D
file-tagger-main$ tagger.sh add -n B -p D
file-tagger-main$ tagger.sh add -n E -pl <p_list_file.txt>
file-tagger-main$ tagger.sh add -n B -p C
file-tagger-main$ tagger.sh remove -n E
Are you sure you want to use the remove command [Y/N]: Y
Establish new relationships with the following parent-child pairs:
Child_node parent_node
G          F
G          X
```

\$vi p_list_file.txt

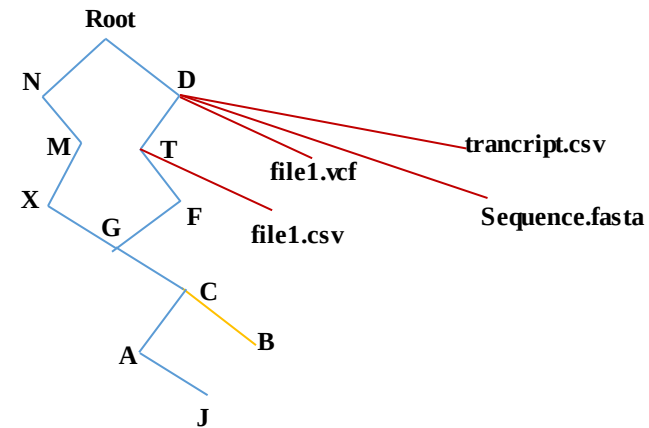
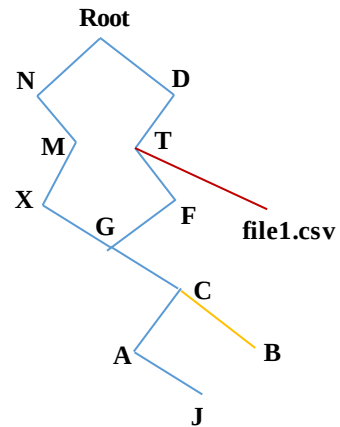
X
F

View node Hierarchy

```
file-tagger-main$ tagger.sh show -n T
Node T Hierarchy
T
F
E
G
C
B
A
J
Files under the T hierarchy
[Error: ] File does not exist on the hierarchy list
```

Tools Commands

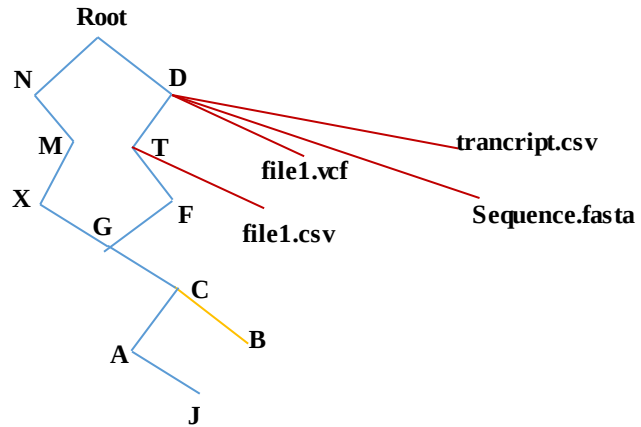
Tag single file and multiple files



```
cuser@cuser-HP-Compaq-6000-Pro-MT-PC: ~/file_tagger-main
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ ls
conn.sh  file1.csv  file1.vcf  file_list  function.sh  init.sql  node_list  README.md  tagger.sh
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ sudo ./tagger.sh add -f file1.csv -p T
added FILE: file1.csv with PARENT-NODE: T
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ cat file_list
/home/cuser/transcript.csv
../data/sequence.fasta
file1.vcf
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ sudo ./tagger.sh add -fl file_list -p D
added FILE: transcript.csv with PARENT-NODE: D
added FILE: sequence.fasta with PARENT-NODE: D
added FILE: file1.vcf with PARENT-NODE: D
```

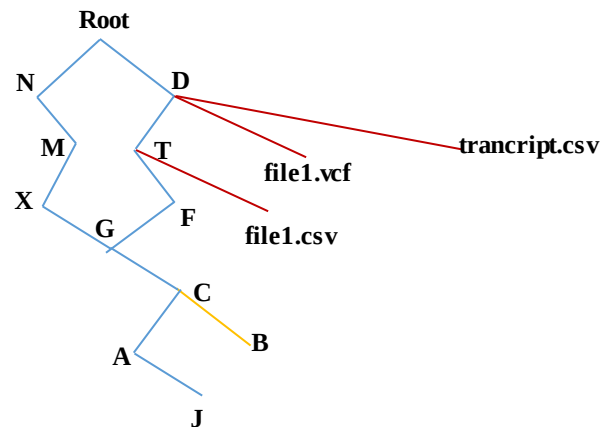
Tool Commands

View files



```
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ sudo ./tagger.sh show -n D
Node D Hierarchy
D
T
F
G
C
A
B
J
Files under D hierarchy
file1.csv      /home/cuser/file_tagger-main/file1.csv
trancrypt.csv  /home/cuser/transcript.csv
sequence.fasta /home/cuser/data/sequence.fasta
file1.vcf      /home/cuser/file_tagger-main/file1.vcf
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$
```

Remove file



```
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ sudo ./tagger.sh remove -f sequence.fasta
Are you sure you want to use remove command [Y/N]: y
Removed File: sequence.fasta
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ sudo ./tagger.sh show -n D
Node D Hierarchy
D
T
F
G
C
A
B
J
Files under D hierarchy
file1.csv      /home/cuser/file_tagger-main/file1.csv
trancrypt.csv  /home/cuser/transcript.csv
file1.vcf      /home/cuser/file_tagger-main/file1.vcf
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$
```

MySQL Tables

```
mysql> select * from tagger_node_node order by id;
```

id	c_node	p_node
2	N	Root
3	D	Root
4	M	N
5	T	D
6	X	M
7	F	T
11	C	G
12	A	C
13	J	A
15	B	C
16	G	F
17	G	X

```
+-----+  
12 rows in set (0.00 sec)
```

```
mysql> select * from tagger_file_info;
```

file_path_id	file_name	system_path
1	file1.csv	/home/cuser/file_tagger-main/file1.csv
2	transcript.csv	/home/cuser/transcript.csv
4	file1.vcf	/home/cuser/file_tagger-main/file1.vcf

```
+-----+  
3 rows in set (0.00 sec)
```

```
mysql> select * from tagger_file_node;
```

file_path_id	file_n	fp_node
1	file1.csv	T
2	transcript.csv	D
4	file1.vcf	D

```
+-----+  
3 rows in set (0.00 sec)
```

```
mysql>
```

Usage

```
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$  
cuser@cuser-HP-Compaq-6000-Pro-MT-PC:~/file_tagger-main$ ./tagger.sh  
  
Usage: ./tagger.sh <command> [options]  
  
Commands:  
  add      Add nodes or files  
  show     Show node hierarchy  
  remove   Remove nodes or files  
  info     Display file path  
  
Options:  
  -n <c_node>          Specify the child node  
  -pl <parent_list_file> Specify a parent list file  
  -p <p_node>          Specify a parent node  
  -f <file_n>          Specify a file  
  -fl <file_list>       Specify a file list  
  
Example:  
  
-Add Nodes  
  ./tagger.sh add -n <c_node> [-pl <parent_file_list>]  
  ./tagger.sh add -n <c_node> [-p <p_node>]  
  
-Add files  
  ./tagger.sh add -f <file_name> [-p <fp_node>]  
  ./tagger.sh add -fl <file_list> -p <parent_node>  
  ./tagger.sh add -f <file_name> -pl <parent_file_list>  
  ./tagger.sh add -fl <file_list> -pl <parent_file_list>  
  
-Searching Node Hierarchy  
  ./tagger.sh show -n <node_name>  
  ./tagger.sh show -pl <parent_list_file>  
  ./tagger.sh show -f <file_name>          it display the parent node  
  ./tagger.sh info -f <file_name>          it display the path of the file  
  
-Remove files  
  ./tagger.sh remove -f <file_n>  
  ./tagger.sh remove -fl <file_list>  
  
-Remove nodes  
  ./tagger.sh remove -n <file_n>  
  ./tagger.sh remove -pl <file_list>
```

THANK YOU !